

Minor Project

 **Task3:** Malware Detection tool

 **Deadline:** 15th Feb, 11:59 PM

Malware:

Malware is software designed to harm, exploit, or compromise a system.

Example include:

- ➔ Viruses
- ➔ Worms
- ➔ Trojans
- ➔ Ransomware
- ➔ Spyware

Malware Detection Tool:

A malware detection tool scans files, applications, or systems to detect malicious threats. It does this by:

- Checking **file signatures** against a known malware database.
- Monitoring **suspicious behavior** (e.g., unauthorized access).
- Detecting **anomalies in system activity**.

Hash Signatures

A **hash signature** is a unique, fixed-length code generated using a **cryptographic hashing algorithm** (such as SHA-256) from a file's contents. It acts like a **fingerprint** for a file.

Why SHA-256?

- Collision-resistant

- **Highly secure (256-bit encryption)**
- **Widely trusted**
- **Ideal for cryptographic security and malware detection**

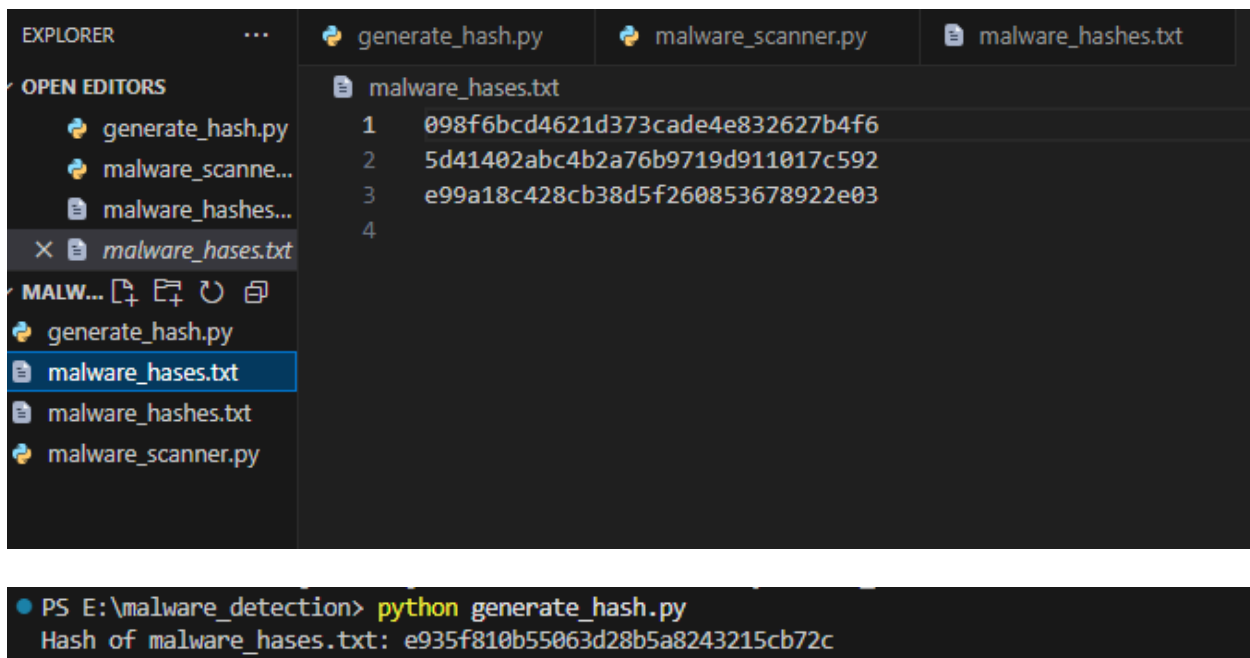
Project structure:

malware_detection

```
| — generate_hash.py    # Script to generate file hashes
| — malware_hashes.txt  # List of known malware hashes
| — malware_scanner.py  # Malware detection scanner
| — README.md          # Documentation
```

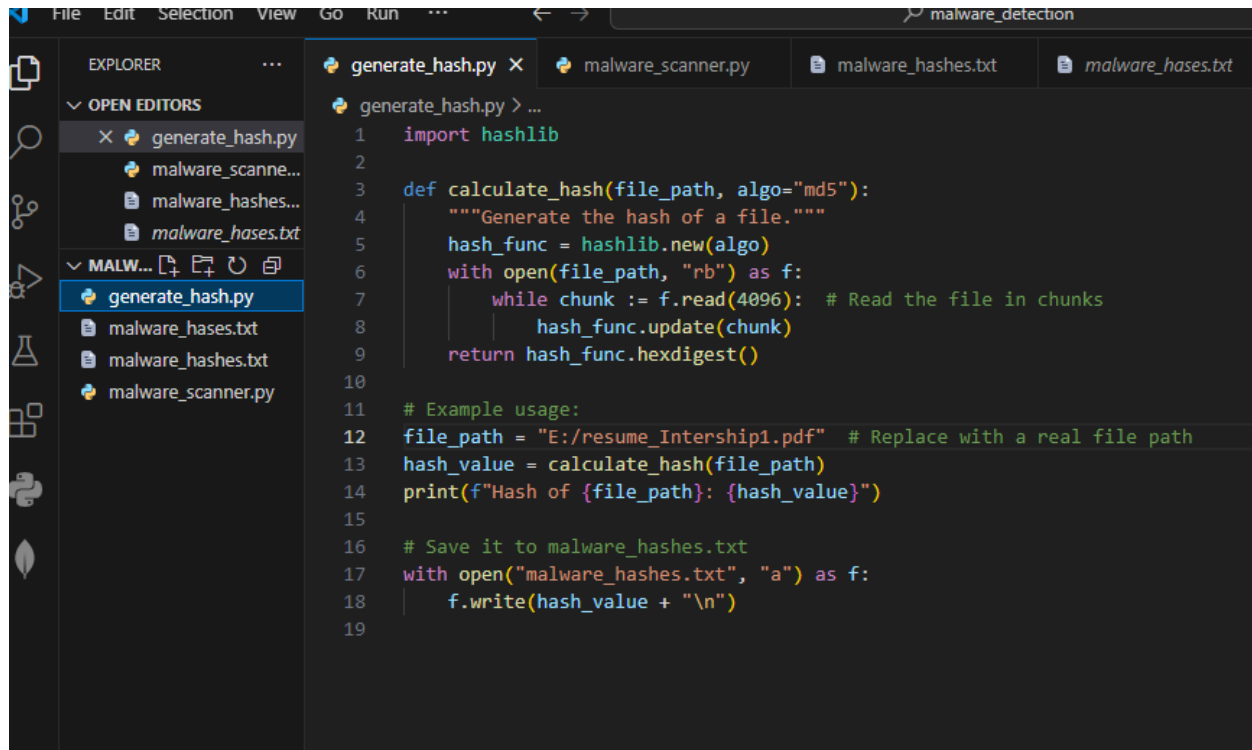
Step 1: creating malware_hases.txt file

The first step is to create a file named `malware_hashes.txt`, which will store hash signatures of known malware files.



Step 2: Generating Hashes for Files:

We created a script `generate_hash.py` to generate SHA-256 hashes for any file

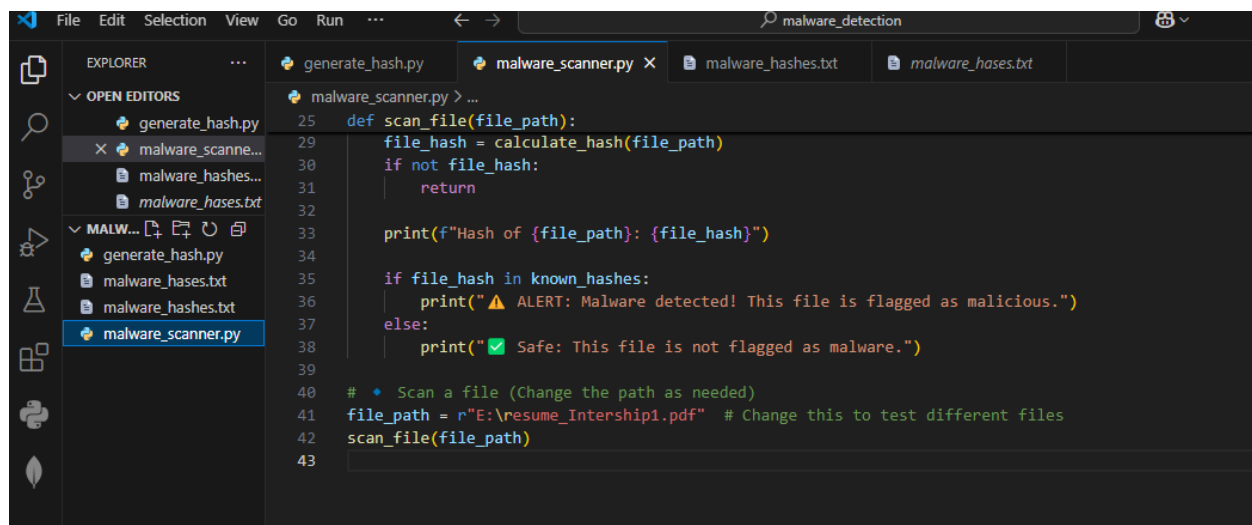


The screenshot shows a code editor with the following files open: `generate_hash.py`, `malware_scanner.py`, `malware_hashes.txt`, and `malware_hases.txt`. The `generate_hash.py` script is displayed in the editor, featuring a `calculate_hash` function that uses `hashlib` to generate hashes for a given file path. It includes an example usage section that calculates the hash for `E:/resume_Intership1.pdf` and saves it to `malware_hashes.txt`.

```
1 import hashlib
2
3 def calculate_hash(file_path, algo="md5"):
4     """Generate the hash of a file."""
5     hash_func = hashlib.new(algo)
6     with open(file_path, "rb") as f:
7         while chunk := f.read(4096): # Read the file in chunks
8             hash_func.update(chunk)
9     return hash_func.hexdigest()
10
11 # Example usage:
12 file_path = "E:/resume_Intership1.pdf" # Replace with a real file path
13 hash_value = calculate_hash(file_path)
14 print(f"Hash of {file_path}: {hash_value}")
15
16 # Save it to malware_hashes.txt
17 with open("malware_hashes.txt", "a") as f:
18     f.write(hash_value + "\n")
19
```

Step 3: Scanning for Malware:

then created `malware_scanner.py`, which compares the hash of a file with those stored in `malware_hashes.txt`.



The screenshot shows the same code editor with the `malware_scanner.py` script now open. The script defines a `scan_file` function that takes a file path, calculates its hash using the `calculate_hash` function from `generate_hash.py`, and compares it against a list of known hashes. If the hash is found in the list, it prints an alert message; otherwise, it prints a safe message. The script also includes a comment and code to scan a specific file path.

```
25 def scan_file(file_path):
26     file_hash = calculate_hash(file_path)
27     if not file_hash:
28         return
29
30     print(f"Hash of {file_path}: {file_hash}")
31
32     if file_hash in known_hashes:
33         print("⚠️ ALERT: Malware detected! This file is flagged as malicious.")
34     else:
35         print("✅ Safe: This file is not flagged as malware.")
36
37 # • Scan a file (Change the path as needed)
38 file_path = r"E:\resume_Intership1.pdf" # Change this to test different files
39 scan_file(file_path)
40
```

Step 4: Running the Malware Scanner

- Run `generate_hash.py` to get the hash of a file.
- Run `malware_scanner.py` to check if the file is malware.

```
PS E:\malware_detection> python generate_hash.py
Hash of E:/resume_Intership1.pdf: 7f155beb76a0663210d7f8d9fed7f566
PS E:\malware_detection> python malware_scanner.py
>>
Hash of E:\resume_Intership1.pdf: 7f155beb76a0663210d7f8d9fed7f566
⚠️ALERT: Malware detected! This file is flagged as malicious.
PS E:\malware_detection> python malware_scanner.py
Hash of E:\resume_Intership1.pdf: 7f155beb76a0663210d7f8d9fed7f566
✅ Safe: This file is not flagged as malware.
PS E:\malware_detection> █
```

☑ Observations and Recommendations

- The scanned system has common web services running (HTTP, HTTPS).
- No immediate vulnerabilities detected, but regular security patches are recommended.
- Secure SSH by restricting access and using key-based authentication.
- Firewall rules should be reviewed to ensure no unnecessary open ports.

🎯 Conclusion

This project successfully implements a **basic malware detection tool** using hash signatures. The scanner helps identify malware by comparing file hashes with a known database of malware hashes.

📄 **Submitted By:** [Deepika Deshmukh]

📅 **Date:** 16th Feb

📁 **Submission:** WEE4 Day 3

